

Lesson 4 — Say the Plan First (Picture Plan)

Matches LEGO Teacher Portal lesson: "Caterpillar Bus".

Objective (I can...): I can tell my plan in words or pictures first, then turn it into blocks. **CS concept:** Planning first / early algorithm planning — saying the plan in plain words is a first taste of **picking the big steps and hiding the little mess.** · **Builds on:** Lesson 3 (little steps).

AI Connection (2 min, read aloud): Grown-ups who make AI say its plan in plain words first — "if it sees a face, wave" — before any code. *Ask the group:* "Why say your plan out loud before you build it?"

Materials: LEGO CS & AI kit + device per pair, a picture "plan mat" per pair, a stack of blank "mission cards." **Model reuse:** same Basics model from Lesson 2 — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Secret mission" — each pair draws or says a 3-step plan on a mission card, then trades cards with another pair. Your card is the other pair's mission to build and run. Clear plans win the mission. - **Concept + demo (8):** Coach says 3 plain-word lines, then matches each to a picture block while kids watch — the plan is the bridge from idea to code. (*Try "guess what it does" with thumbs up/down before Run.*) - **Build/Code (25):** Pairs build the mission card they got: say or draw the plan on the mat, then code it step-by-step on the built model and run to check it matches. **Make-it-yours beat:** add or swap one brick to fit the mission's theme. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change one plan step (like a longer wait) and fix the matching block so plan and code still agree. - **Share + tidy-up (10):** Pairs point to their plan, then run to prove it matches. Model stays built.

Pair roles: Navigator owns the plan mat; Driver turns each line into a block. Swap at the wiggle break. (*Trio: third child checks each plan step got coded.*)

Differentiation: *Support* — a plan mat with picture slots to fill ("First . **Then** ."). *Extension / fast-finisher* — a 5-step plan. *Reteach* — code one plan step together, then let them do the next alone.

Exit check: Pair's plan steps line up one-to-one with their working blocks. **At-home (optional):** Draw a 4-picture plan for getting ready for bed. ~5 min.